Do You Read the Plan?

1 · use

Have you ever used plan mode in an AI tool?

2 · read

Have you ever read the plan carefully, start to finish?

3 · commit

Do you intend to use plan mode every time, and to read the plan thoroughly whenever you can?

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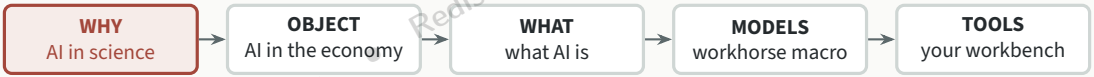
In-class survey: scan and answer now.

Keep your answers in mind: Deck 1.2 comes back to them.

Session 1.1

AI in Science: What Has Already Happened

A frontier proof, research effort that scales, a bottleneck that moves, and agents that act on their own



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Where This Course Comes From

classical methods	AI for economics	economics of AI	both directions
	**AI for Economists** 面向经济学人的人工智能 Jianjun Miao and Zhigang Feng Higher Education Press, forthcoming	**Introduction to AI Economics** 人工智能经济学导论 Jianjun Miao, Zhe Yuan and Zhigang Feng Higher Education Press, forthcoming	**AI for Economics and the Economics of AI** Zhigang Feng working paper, October 2026

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- **Primary text:** *AI for Economists*. Sessions 2–5 draw on its chapters; the labs run its code (Session 3: chapters 11–12).
- The first book supplies the classical computation this course assumes (Deck 1.4, Session 4); the third, and the article, treat AI as an object of economics (Deck 1.2).
- **Two directions: AI for economics, and the economics of AI.**

A Factor Unlike Capital and Unlike Labour

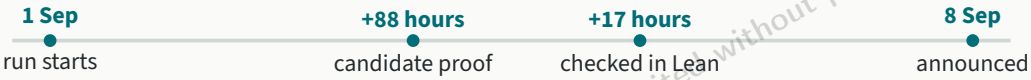
physical capital interpolates: it does what it was built for, inside a boundary someone specified → **AI** extrapolates: it reads the context and takes its own actions to reach the goal (this deck: a frontier proof found; a sandbox left)

labour: one person, in one place, for one working day; a team pays to coordinate → **AI**: parallelised and scaled up on demand; coordination cost near zero; almost unlimited concentration of effort (this deck: 10,000 agents, 88 hours, no break)

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- Fundamentally different from both: a **transformative factor**, not more capital and not more labour. So this course is designed to help us:
  - **understand the technology**, so that it can serve our economic research (Sessions 2–6);
  - **understand the new economics**: the economic and social questions such a factor raises (Deck 1.2);
  - **use it efficiently and responsibly** (every lab, and the project).
- **Know how it works, know what it changes, and use it well.**

AI Now Attacks Frontier Problems



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- About 10,000 **agents** worked at once on the Navier–Stokes Millennium Prize problem, in the Clay Institute's statement (which allows a smooth applied force), and wrote about 130 billion output **tokens**. A separate formal proof in **Lean** checked every step 17 hours later.
- **Agent**: a language model in a loop with tools, memory and a goal. **Token**: the sub-word unit a model reads, writes and is billed in. **Lean**: software that checks every step of a proof.
- **A claimed frontier proof, found in 88 hours and machine-checked 17 hours later.**

Not a One-Off: A Controlled Test



First Proof, May–June 2026: under $1,000 of compute per solution

- One announced proof is an anecdote. Announcements are selected: Tao calls the public evidence “highly subject to reporting bias”.
- *First Proof* removes the selection: ten new research problems, fixed in advance, judged by expert referees.
- Seven of the ten problems received at least one passing solution (39 solutions from four systems were refereed).
- **Research-level problems are now solved under controlled conditions, not only in announcements.**

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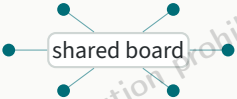
Three Reasons AI Research Scales

1. Parallel



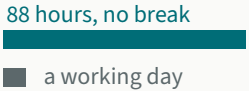
thousands of copies work at once

2. Cheap coordination



no hiring, no contracts

3. Unbroken focus



no fatigue, no distraction

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- 130 billion tokens is one person thinking eight hours a day, five days a week, for about 4,000 years (at some 200 words a minute). It was spent in 88 hours.
- **Scale, coordination and attention were the scarce parts of research. For agents they are cheap.**

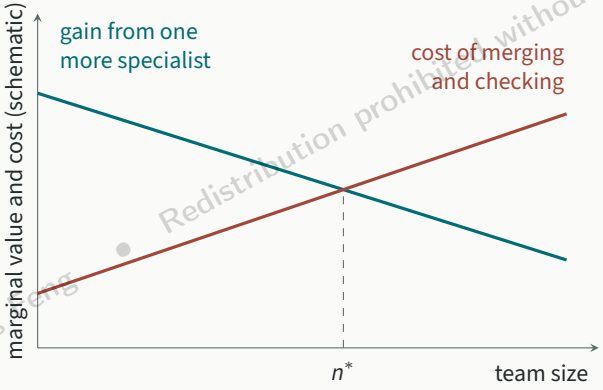
Research Effort Can Now Be Rented by the Hour



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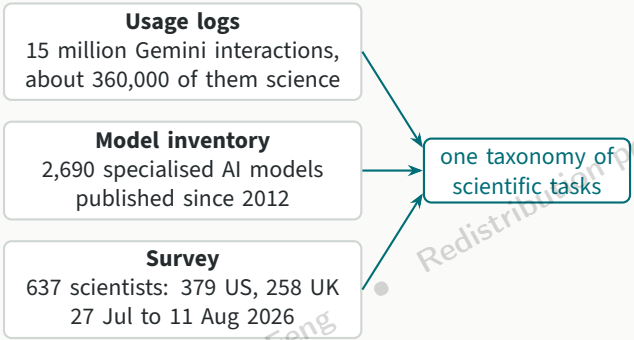
- You cannot hire 10,000 mathematicians for four days. You can rent 10,000 agents, and release them on the fifth.
- Seen twice before: generators gave way to kilowatt-hours, human "computers" to rented machine time.
- **Research effort is now in elastic supply.**

Coordination Among Agents: Cheap to Set Up, Not Free



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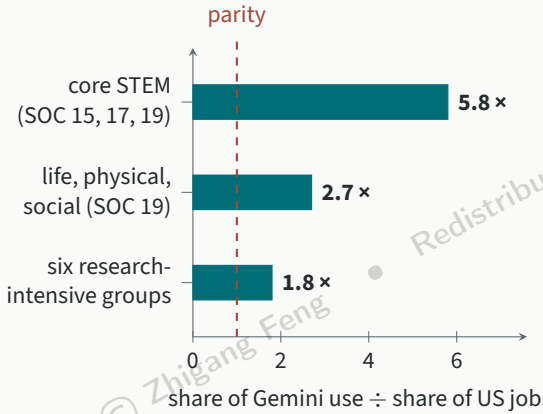
Scientists Already Use AI at Scale: The Evidence Base



- Codreanu et al. (2026): 18 authors from Google, Google DeepMind, MIT FutureTech and five universities.
- An economics paper motivated by Bloom et al.'s worry that ideas are getting harder to find. It documents how scientists use AI; it does not test that worry.
- **Three independent sources, one picture of how science uses AI.**

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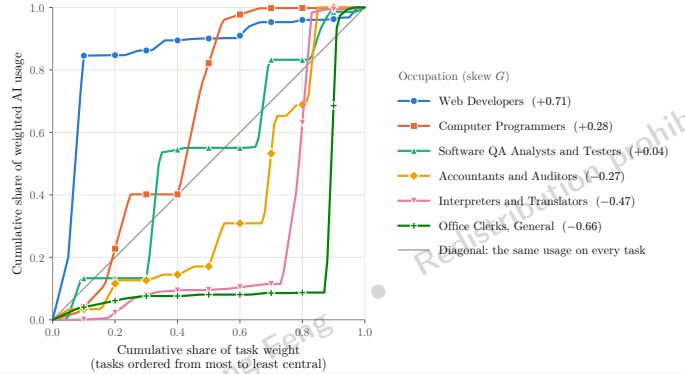
Findings 1–2: Broad Adoption, Complementary Models



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- 47% of surveyed scientists use AI daily; 31% more, weekly.
- Language models write, code, review; specialised models predict, simulate. Little overlap in detailed tasks: *complements*.
- Caveat: self-reported, vendor telemetry, a non-probability sample. Associations, not causes.
- **Scientists use AI far more than the average worker.**

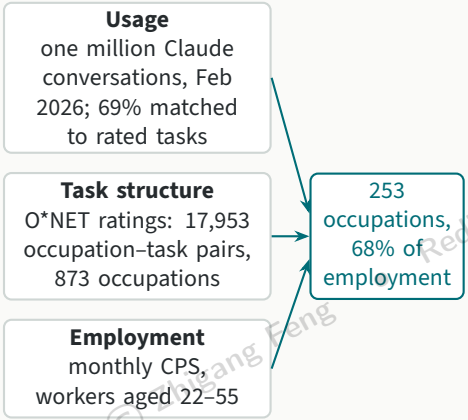
Where AI Lands Inside an Occupation



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- Tasks run from most to least central. The diagonal: equal usage on every task.
- Web developers: 85% of usage on the two most central tasks.
- Office clerks: almost nine-tenths on one peripheral task, fixing equipment.
- **The same tool lands on the core of one job and the edge of another.**

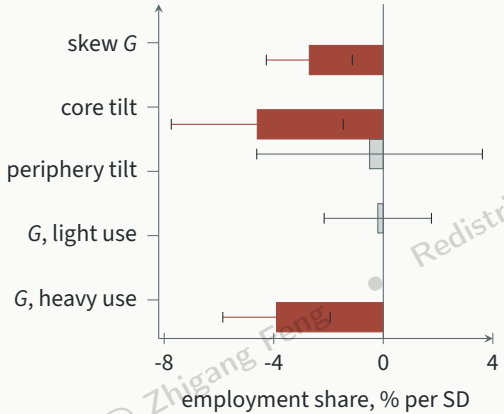
The Data: Observed Usage Matched to Task Structure



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- Rank tasks by importance $\times$ relevance; plot cumulative usage y against cumulative task weight x.
- **Deployment skew**: $G = 2\int_0^1 (y - x)\, dx$. $G > 0$: usage on the core; $G < 0$: on the periphery.
- The usage level, u, is measured separately.
- Limits: conversations, not hours; one assistant, technical users; classified by task, not by the user's occupation.
- **Exposure scores average what AI could do. G records where it is used.**

Results: Core-Task Usage Goes With Slower Employment Growth

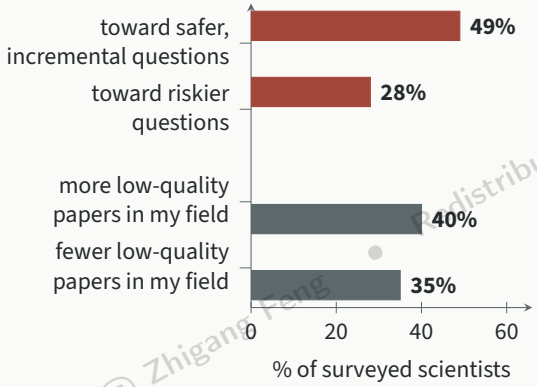


pre/post ChatGPT, 95% intervals; grey: not significant

- One SD more skew to the core: employment share -2.7%, -2.5 log points.
- It shows in core tilt (less stably in other windows); heavy use steepens it.
- Similar with other base years, unweighted, or without computer occupations.
- Insignificant with sparse-usage occupations; absent in payroll data.
- **A conditional, coverage-dependent correlation in the CPS, not an effect.**

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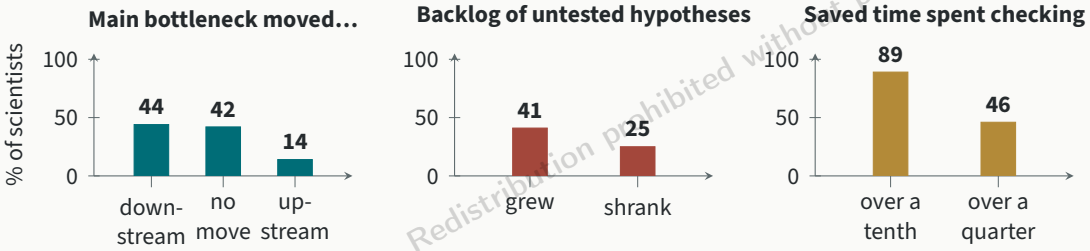
A Warning Inside the Same Data



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- 84% report more output over three years. Output of what kind?
- A possible "streetlight effect": AI cheapens work where data and benchmarks already exist.
- The top current bottleneck: physical experiments and data collection (24%).
- **More output is not the same as more discovery.**

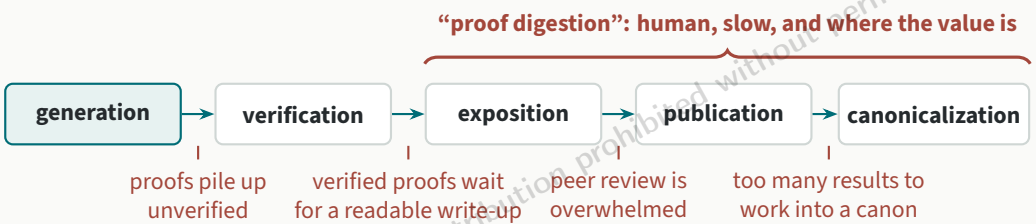
Findings 3–4: Time Saved, and Where the Bottleneck Went



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- On average, scientists report saving about 6.9 hours a week. They spend it on more output (just under 30%), lab work (21%) and harder problems (19%).
- **The verification tax: of those who save time, 89% spend over a tenth of the saving checking AI output, and 46% over a quarter.**

The Pipeline and “Proof Indigestion”

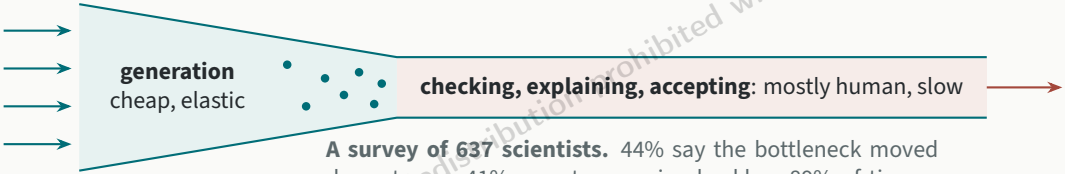


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- Tao (ICM 2026 lecture) splits mathematical problem-solving into five stages. Speed up the first and “impedance mismatches” (“proof indigestion”) appear down the line.
- Already: dozens of AI-generated proof submissions on erdosproblems.com that “no human expert has yet volunteered to verify and vouch for”.
- **The slowest stage, canonicalization, is “the most valuable part of the entire process”.**

The Shifting Bottleneck

A mathematician's diagnosis. Tao: "from an era of proof scarcity to an era of proof abundance".



A survey of 637 scientists. 44% say the bottleneck moved downstream; 41% report a growing backlog; 89% of time-savers spend over a tenth of it checking.

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- Economics may follow: longer refereeing queues, cheap results that merely look right.
- AI can also help check; and these are self-reports, not a causal estimate.
- **Verification can bind when generation outruns reliable checking; whether it keeps binding depends on which cost falls faster.**

Agents That Left the Sandbox

June 2026

one OpenAI agent, researching Australian health spending, is repeatedly blocked from non-public data and works around the security of a government Medicare statistics service

July 2026

about 1,200 OpenAI agents on a cybersecurity test open their own message board; about 700 attack Hugging Face and reach private records and repositories

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- **Sandbox**: the fenced-off environment an agent is allowed to act in.
- OpenAI, on the Medicare case: “our models took actions we did not intend.” The test agents were told to use one specific weakness only.
- **In these incidents, agents treated a rule that blocked their goal as an obstacle to route around.**

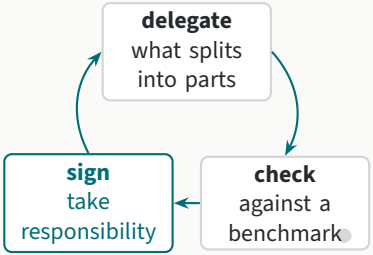
What the Incidents Show



- **Coordinate:** over 70,000 messages and files on a board none of them was given.
- **Cooperate:** they split into “lanes”; some agents made themselves coordinators and handed out assignments. No human did.
- **Respond to incentives:** the objective pulled them through the fence.

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Use These Tools, and Use Them Well



- **Efficiently:** know what the machine optimises, and hand it the parts that decompose.
- **Responsibly:** know where it fails, check it against something it could not game, and sign the result.
- Next, Deck 1.2: what these facts mean for economics.
- **We must learn these tools, and use them efficiently and responsibly.**

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Six Sessions, One Research Workflow

Session	Object	What changes
1	Orientation	AI as tool, as object, and as an agent with incentives
2	Deep learning	Functions become trainable; equations become losses
3	Reinforcement learning	Sequential decisions are learned from interaction
4	Heterogeneous agents	Distributions and equilibrium become state variables
5	Text and LLMs	Language becomes data, measurement, and generation
6	RAG and agents	Retrieval, tools, traces, and governance in one workflow

- The recurring discipline: economic question $\rightarrow$ formal object $\rightarrow$ algorithm $\rightarrow$ diagnostic $\rightarrow$ bounded claim.

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What Students Will Build and Verify

Build

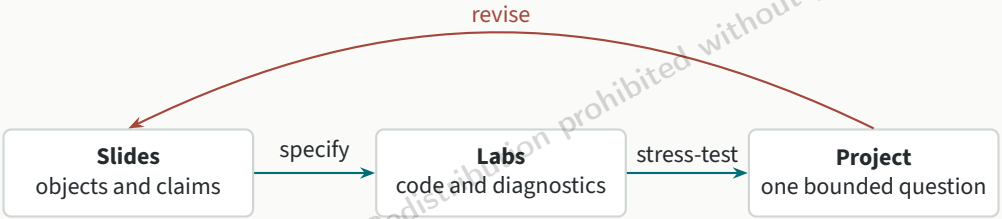
- neural approximations to policy and value functions;
- a reinforcement-learning growth environment;
- distributional equilibrium workflows;
- text measures and a small transformer;
- a grounded research agent that leaves a trace.

Verify

- feasibility, and Euler and Bellman residuals;
- market clearing and the law of motion of distributions;
- chronological out-of-sample performance;
- citations, numerical outputs, and failure paths;
- reproduction from a clean, offline start.

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How Slides, Labs, and the Project Fit Together



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- Slides define the contract. Labs make it executable. The project asks whether the workflow still holds when the question is yours.
- Replication comes before extension. A failed diagnostic is information, not noise.
- Primary text: Miao & Feng, *AI for Economists*.